MILITARY SPECIFICATION

AMPLIFIER, ERROR LEVEL, ASSEMBLY

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement of Error Level Amplifier Assembly, BUOORD LD 498332. This amplifier assembly is a component of the Fire Control System Mark l12 Mod 1.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation for bids, or as noted, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-P-116	**Preservation, Methods of**
MIL-E-5272	**Environmental Testing, Aeronautical and Associated Equipment, General Specification for**
MIL-T-17113	**Tests, Shock, Vibration, and Inclination (for Electronic Equipment) General Specification**
MIL-T-l8870	**Fire Control, Optical and Associated Equipment Intended for Shipboard Use, The General Specification for Design and Procurement of**

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STANDARDS

MILITARY

MIL-STD-l05	Sampling **Procedures and Tables for Inspection by Attributes**
MIL-STD-167	**Mechanical Vibrations of Shipboard Equipment**

DRAWINGS

NAVY DEPARTMENT

BUORD LD 498332	Error Level Amplifier Assembly
BUORD SA 2530410	Mark 112 Amplifier Test Fixture
BUWEPS 2184910 (FSN A8140-789-2995)	Container, shipping and storage

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVWEPS OD 17184	Description and Operating Instructions for Mark 112 Amplifier Test Fixture Shop Accessary SA 2530410
NAVORD OP 400	General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment
NAVORD OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, and drawings requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The followlng documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28 and all supplements thereto	Screw-thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, 111.)

AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W. Washington 6, D.C.)

3. REQUIREMENTS

3.1 Description. The Error Level Amplifier Assembly, BUORD LD 498332, IS a dual input error sensing device. When the magnitude of either error signal exceeds the threshold level of the unit, an external relay is actuated which reduces the error level sufficiently to assure proper functioning of the associated servo loops.

3.1.1 Design. The product covered by this specification shall be capable of meeting the requirements of the documents listed on drawings of BUORD LD 498322 and MIL-F-18870.

3.2 General requirements.

3.2.1 Preproduction samples. When specified in contract or order, preproduction samples of the Error Level Amplifier Assembly shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. Provision of samples shall be in accordance with 4.1.2.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and parts entering into the final

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assembly of the Error Level Amplifier Assembly shall conform to the detailed requirements indicated on the drawings and specifications listed in BUORD LD) 498332.

3.2.3 Finishes, coatings, construction and dimensions. Finishes, coatings, construction and dimensions for the Error Level Amplifier Assembly shall be in accordance with the applicable drawings, specifications, and publications listed in BUORD LD 498332.

3.2.4 Marking. Markings of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in BUORD LD 498332.

3.2.5 Interchangeability. **Component parts of the assembled units fabricated in strict accordance with their applicable drawings shall be interchangeable and shall assemble with mating components of the units without selection or fitting. If, during the assembly of the fabricated parts of a unit, selection or fitting is required, then changes in dimensions are necessary and revision and correction of the applicable drawings are required. See 3.2.9.**

3.2.6 Gages. . The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for information and guidance in design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responcibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages. (For test equipments, see 4.3.)

3.2.7 Threads. Unless otherwise specified, all threads shall be In accordance with the National Bureau of Standards Handbook H28 and all supplements thereto. The class or fit for threads ahall be as specified on the applicable drawings and the NOT-GO gaging practice shall be as set forth in the supplement to the aforementioned Handbook H28.

3.2.8 Requests for deviation. Requests for deviation from this specification, applicable drawings, specifications, publications, materials, and processes specified herein, shall be as set forth in OP 400.

3.2.9 Conflicting requirements. Conflicting requirements arising between this specification and any specification, publication or drawing

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listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.3 Definitions. Definitions applicable to the environmental requirements of this specification are as follows:

3.3.1 Tropical environment. Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure which present an opportunity for fungi growth.

3.4 Performance requirements and product characteristics. The Error Level Amplifier Assembly shall meet the following performance requirements and product characteristics:

3.4.1 Threshold accuracy.

3.4.1.1 Input signal "R-P."

3.4.1.1.1 When tested with the special test equipment specified in paragraph 4.3.1, the threshold response of the amplifier to a slowly increasing signal applied at pins "R" and "P" shall be such that when the "Relay Energized" light, located on the special test equipment panel, is first illuminated, the input signal as measured by the vacuum tube voltmeter specified in paragraph 4.3.2 shall be between 5.0 and 7.0 volts alternating current.

3.4.1.1.2 When the input signal is increased from that given in 3.4.1.1.1 to 63volts, the "Relay Energized"light shall remain lit.

3.4.1.1.3 When the input signal is slowly decreased until the "Relay Energized" light is extinguished, the external voltmeter shall indicate no less than 3.0 volts.

3.4.1.2 Input signal "M-P."

3.4.1.2.1 When tested with the special test equipment specified in paragraph 4.3.1, the threshold response of the amplifier to a slowly increasing signal applied at pins "M" and "P" shall be such that when the "Relay Energized" light, located on the special test equipment panel, is first illuminated, the input signal as measured by the vacuum tube voltmeter specified In paragraph 4.3.2 shall be between 5.0 and 7.0 volts alternating current.

3.4.1.2.2 When the input signal is increased from that given in 3.4.1.2.1 to 63 volts, the "Relay Energized"light shall remain lit.

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3.4.1.2.3 When the input signal is slosly decreased until the "Relay Energized" light is extinguished the external voltmeter shall indicate no less than 3.0 volts.

3.5 Environmental requirements. The Error Level Amplifier Assembly shall meet the other applicable requirements of this specification during and after tests defined or specified in 4.6.

3.5.1 Vibration, operating condition. The Error Level Amplifier Assembly shall be capable of withstanding the Type I vibration requirements as specified in MIL-D-167 and shall meet all performance and product characteristics requirements of this specification during and after vibration tests and thereafter show no damage.

3.5.2 Shock, operating condition. The Error Level Amplifies Assembly shall be capable of withstanding the Class A shock requirements as specified in MIL-T-17113 and shall meet all performance and product characteristics requirements of this specification before and after shock tests and thereafter shall show no damage.

3.5.3 Temperature. The Error Level Amplifier Assembly shall be capable of withstanding operational ambient temperatures of between zero degrees Fahrenheit (minus 18 degrees centigrade) and 185 degrees Fahrenheit (85 degrees centigrade) in accordance with MIL-F-18870 and storage temperatures of between minus 40 degrees Fahrenheit (minus 40 degrees centigrade) and 160 degrees Fahrenheit (71 degrees centigrade) in accordance with Condition II of NAVORD OS 6341, and thereafter shall show no corrosion or damage and shall meet all the performance and product characteristics of this specification.

3.5.4 Humidity. The Error Level Amplifier Assembly shall be capable of withstanding operational ambient humidities in accordance with MIL-F-18870 and thereafter shall show no corrosion or damage and shall meet the performance and product characteristics of this specification.

3.5.5 Tropical environment. The Error Level Amplifier Assembly shall be capable of withstanding the tropical environmental conditions as specified in MIL-E-5272 under 'Fungus Resistance Tests" and thereafter shall show no fungus growth, corrosion *or* other damage, and shall meet the performance and product characteristics of this specification.

3.6 clean-up. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

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3.7 Workmanship. **The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, marking of parts and assemblies, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges.**

3.8 packaging, packing and marking. Packaging, packing and marking shall be as specified In Section 5 and meet the test requirements specified in Section 4 of MIL-P-116.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures **used by Government inspectors in the determination of the acceptability of products submitted by a supplier for Government inspection, shall be in accordance with the provisions set forth in MIL-STD-105.**

4.1.2 Preproduction sample. When specified for each contract or order, three (3) Error Level Amplifier Assemblies shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for preproduction tests at an activity designated in the contract or by the Bureau of Naval Weapons for purposes of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 **Lot sizes. As applied to Government inspection of units of product, the term 'lot" shall mean 'inspection lot." i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product** in **"inspection lots" will be as determined by the Government inspector** and may differ from the quantity designated in the contract or order as a lot

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for production, shipment, or other purpose.

4.1.4 Periodic production samples. From each succeeding lot of 25 or less, a random sample of (1) Error Level Amplifier Assembly preserved and packaged as specified by the contract or order, may be selected by the Government inspector for submission to a Bureau of Naval Weapons assigned activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The Error Level Amplifier Assembly shall be considered satisfactory if the Error Level Amplifier Assembly meets all specifications and pertinent drawing requirements and withstands the tests as specified in 4.5 and 4.6 inclusive.

4.2 Classification of test. The inspection and testing of the Error Level Amplifier Assembly shall be classified as follows:

4.2.1 Acceptance tests. These tests are to be accomplished on each Error Level Amplifier Assembly being submitted for acceptance under contract. Acceptance tests shall be either performed by the manufacturer and witnessed by the Government inspector, or these tests may be conducted by the Government inspector. These tests are detailed in 4.4.

4.2.2 Environmental tests. Environmental tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.4.4, which are representative of the production of the Error Level Amplifier Assembly after the award of the contract to determine that the production meets the requirements of this specification. These are detailed in 4.5. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot, or the cessation of production as determined by the procuring activity.

4.2.3 Tests after delivery. Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein after delivery. Any defects discovered during or after these teats due to the contractors failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.3.1 Special test equipment. Shop Accessory SA 2530410 Mark

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112 Amplifier Test Fixture.

4.3.2 Standard test equipment. One alternating current vacuum tube voltmeter capable of reading voltages from 1 to 10 volts, and 10 to 100 volts. The input impedance shall be 500,000 ohms minimum at 400 cycles per second. The accuracy shall be 2 per cent at fill scale deflection.

4.4 Acceptance teats.

4.4.1 The contractor or Government Inspector shall conduct acceptance tests as specified in 4.2.1 to assure that the Error Level Amplifier Assembly is in compliance with the requirements of 3.2, 3.4, 3.6, and 3.7.

4.4.2 Test conditions. Unless otherwise specified, the Error Level Amplifier Assembly shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature. Room ambient 65 degrees Fahrenheit to 95 degrees Fahrenheit (18 degrees centigrade to 35 degrees centigrade).

4.4.2.2 Altitude. Normal ground atmospheric pressure.

4.4.2.3 Vibration. None.

4.4.2.4 Humidity. Room ambient to 95 per cent relative maximum.

4.4.3 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary standard and special test equipment and facilities and personnel for performing all acceptance tests. If specified in the contract, the Government may furnish special test equipment specified in 4.3.1 when it is available. The standard and special test equipment shall be adequate in quantity and, unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure. Follow the detailed test procedure given in NAVWEPS OD 17184 'Description and Operating Instructions for Marls 112 Amplifier Test Fixture Shop Accessory SA 2530410" and test the amplifier to determine compliance with requirements of paragraph 3.4.

4.5 Environmental tests.

4.5.1 Vibration, operating condition. The Error Level Amplifier Assembly rigidly mounted in a suitable vibration fixture, shall be subjetted to the Type I vibration tests as specified in MIL-STD-167 to

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determine compliance with requirements specified in 3.5.1 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.2 Shock, operating condition. The Error Level Amplifier Assembly rigidly mounted in a suitable shock fixture, shall be subjected to Class A shock tests as specified in MIL-T-17113 to determine compliance with requirements specified in 3.5.2 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3 Temperate and humidity. The Error Level Amplifier Assembly shall be subjected to operational ambient temperature and humidity tests specified in MIL-F-18870 and storage temperatures in accordance with Condition II of OS 6341 to determine compliance with requirements specified in 3.5.3 and 3.5.4 and thereafter shall meet the Performance requirements and product characteristics of this specification.

4.5.4 Tropical environment. The Error Level Amplifier Assembly shall be exposed to Fungus Resistance Tests specified in MIL-E-5272 to determine compliance with requirements specified in 3.5.5 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6 Acceptance criteria. Sample units which fail to comply with any of the tests or inspection requirements shall be classified as defective. If the number of defective units exceeds the number of defective permitted by the associated acceptance criteria, the lot represented will be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packa ing. Preservation and packaging shall be Level A, B or C as specified (see 6.2.)

5.1.1 Level A

5.1.1.1 Cleaning. Error Level Amplifier Assembly shall be cleaned in accordance with Method C-1 of Specification MIL-P-116.

5.1.1.2 Drying. Error Level Amplifier Assembly shall be dried by one or more of the drying procedures listed in Specification MIL-P-116 provided that the procedure(s) selected will not cause damage to the assembly.

5.1.1.3 Preservative application. Not applicable.

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5.1.1.4 Unit packaging. Error Level Amplifier Assembly shall be packaged in accordance with Method IId of Specification MIL-P-116 in a reusable oontainer conforming to BUWEPS drawing 2184910, Federal Stock Number A814-789-2995.

5.1.1.5 Intermediate Packaging. Not applicable.

5.1.2 Levels B and C. Not applicable.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2)

5.2.1 Level A. The reusable container specified in paragraph 5.1.1.4 is an exterior type shipping container, therefore additional packing for shipment is not required.

5.2.2 Levels B and C. Not applicable.

5.3 Marking. Marking shall be as specified on BUWEPS drawing 2184910.

6. NOTES

6.1 Intended use. The Error Level Amplifier Assembly is intended to be used in the Mark 112 Mod 1 Fire Control System.

6.2 Ordering data. Procurement documents should specify the following:

(a) The specific title, number and date of this specification.
(b) Selections of applicable levels of preservation, packaging and packing.
(c) "The exceptions to this specificatlon, applicable drawings and other documents.
(d) The guarantee period (see 4.2.3).

6.3 Explanation of Level A packaging and packing.

6.3.1 Level A—Military package. Level A-Military package is the preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during shipment~ handling, indeterminate storage, and worldwide distribution.

6.3.2 Level A—Military pack. Level A-Military pack isa packing which will afford adequate protection during shipment, handling, indeterminate storage, and worldwide distribution.

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